

Instruction

1 General information

Desiccant filling on-site instruction (9829226201)

Applicable to:	Desiccant filling on-site	
Safety Instruction:	General	
Revision:	Date:	Modified from:
1	07.03.2023	Heavy duty of 3600-4150 added; Desiccant filling of 6000-10000 added;
2	11.09.2023	Section "Package, Transport and Storage" added; Std & Heavy filling of XD2750 ZP G added; Desiccant ratio of 3600/4150/5000 modified;

2 Document Overview

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3 Package, Transport and Storage

1. The desiccant shipping packages adopt double-layer packaging, the inner bag is a polyethylene film bag, and the outer bag is a plastic woven bag or a paper-plastic-composite packaging bag. The inner bag is tied or heat sealed, and the outer bag should be firmly sewn tightly.

If the user has special requirements, the supply and demand parties shall negotiate.

2. During transportation and storage, the desiccant should be avoided from mixing with volatile substances, should be protected from rain and damage. The packaging should be handled with care to avoid contamination.

3. The shipping packages with desiccant should be stored in a dry, enclosed warehouse. These packages must be protected from harm from nature and surroundings.

4. The shipping packages with desiccant must be stored in an area that is off the ground and well drained to prevent contact with accumulated water or mud.

5. The shipping packages should be covered with waterproof or oilcloth to prevent water accumulation or infiltration at the top.

6. The shipping packages with desiccant are not recommended for long-term storage outdoors. The impacts of outdoor storage on the desiccant are unknown.

7. If shipping packages with desiccant do need to be stored outdoors for a short period of time, measures should be taken to prevent rain, water, and moisture, as well as UV (ultraviolet) degradation.

8. The shelf life of desiccant supplied through large bag packaging varies depending on the type of product, so please consult your local representative for the storage time. It is advised that desiccant shall be filled into the specific pressure vessel as soon as possible to protect desiccant and ensure its performance.

9. Due to the strong adsorption capacity of desiccant, if the storage time is too long, the desiccant should absorb moisture from the air, which will affect the adsorption performance. If the situation is serious, the desiccant needs to be regenerated for several cycles before it can enter a good working state.

10. Any damage or malfunction caused by not following the above instructions is not covered by Warranty or Product Liability.

11. Package, transport and storage of desiccant should comply with relevant laws and local regulations. Activated alumina refers to HG / T 3927-2020. Silica gel refers to HG/T2765-2005. Molecular sieve refers to HG/T2524-2010. For more details, contact your local representative.

4 Safety

4.1 Safety first

SAFETY = THINK BEFORE YOU ACT



1. STOP
EVALUATE RISKS



3. THINK
SEEK PROPER SOLUTION



2. ACT
CARRY OUT SOLUTION



KNOW THE SAFETY RULES OF YOUR CUSTOMER

ALL VISITORS MUST REPORT TO RECEPTION



KNOW THE SAFETY INSTRUCTIONS OF YOUR MACHINE



PERSONAL PROTECTION KEEPS YOU SAFE



LOCK OUT / TAG OUT
TAKE AWAY THE RISKS

4.2 Prepare work

Attention: Protect yourself, protect others. Reading and understanding this section completely before filling the desiccant and pay attention to all warning labels for shipping packages.

When the desiccant is fresh and unused, it's property is not flammable. However, when it's exposed to water, it's temperature could be raised quite high. When the desiccant first time become wet, the temperature can reach the boiling point of water. Therefore, the desiccant should not be put in the mouth, or contact with the eyes.

Desiccant dust may irritate your nose, throat, eyes, lungs and skin. As some desiccant dust will be produced when filling with fresh desiccant, it is necessary to wear protective glasses, dust mask, gloves and protective clothing.

Desiccant is usually stored within a bag or barrel package with a specified weight. Pay attention to safety when opening the cover of the bag package, you must wear protective glasses, dust mask, gloves and protective clothing for protection.

Since desiccant filling port is at the top of the vessel, working at high altitude above ground is needed, therefore local climbing operation safety regulations must be observed.

4.2.1 Evacuation of desiccant filling area

Due to all dryers desiccant filling port are at the top of the vessel, the desiccant must be lifted to the top of the vessel by the crane, therefore operating area of the crane must be identified, all personnel must be kept away from the crane operation area and stand within the safety area. Sometimes a pneumatic filling system is used to blow the desiccant into the vessel through the pipe connection. The inlet and outlet of the pipe should be kept away from people because of the air pressure in the storage tank. As the desiccant passes through the system, the transportation and receiving system must be well grounded to eliminate the static charge. If flammable vapors, such as hydrocarbons, are present in the filling area, well grounding is extremely important to avoid electrostatic explosions and fire.

4.2.2 Personnel protection equipment (PPE)

Operator must wear safety belts and ropes when entering the vessel.

If operators must enter the vessel, the following precautions must be taken to prevent suffocation:

1. A protection plate shall be installed in all inlet and outlet piping to prevent accidental discharge of dangerous fluids into vessel. Before operator enters, an inside air quality analysis must be done to ensure operator work inside the vessel can breathe safely.

2. Keep air well ventilated inside the vessel. The compressed air cannot be used for ventilation, since compressed air will easily be contaminated or filled by other gases due to faulty operation.

3. Operator who work at high altitude must be wear suitable PPE and observe local climbing operation safety regulation.

Only well trained and PPE equipped operator can be entered the vessel. Make sure operator always wear seat belts and safety ropes. Another two well trained and PPE equipped operators must keep an eye on the operators whom work inside the vessel.

4.2.3 Safety requirement

(1) Wear PPE

Gloves, Protective glasses, Dust Mask, Safety shoes, Safety vest.

Warning: Operators must wear safety ropes and seat belts when working high above the ground.

- (2) Avoid the dusting of desiccant particles;
 - (a) Slowly open the desiccant transportation package;
 - (b) When the desiccant is filled by a pneumatic filling system, the outlet shall not toward to the operator.
- (3) Eliminate the potential risk of the desiccant bag fallen off – nobody can walk into the crane lifting area. Before working on lift the desiccant bag by crane, operating area of the crane must be identified, all personnel must be kept away from the crane operation area and stand within the safety area;
- (4) If the desiccant is filling by a pneumatic filling system, avoid flammable steam burning, pneumatic filling system must be well grounded to avoid static electricity and cause fire. A static spark can ignite any flammable vapor at the site;
- (5) Avoid the possibility of suffocation and falling before entering the vessel or working at the top of the vessel:
 - (a) Install baffles at the inlet and outlet of the vessel;
 - (b) Measure the air quality inside the vessel to ensure safety access;
 - (c) Operator must be well trained and PPE equipped , also safety rope is needed;
 - (d) Another two well trained and fully PPE equipped operator must be cooperate with the one whom will work inside the vessel or work at the top of the vessel.

5 Information of filling on-site dryer

5.1: Machine type above XD3600 or total weight (with or without desiccant) over 15000kg, it's required to ship out the machine without desiccant, due to safety reason during the transportation. Filling desiccant can only start once the machine has been located at the correct position at customer site.

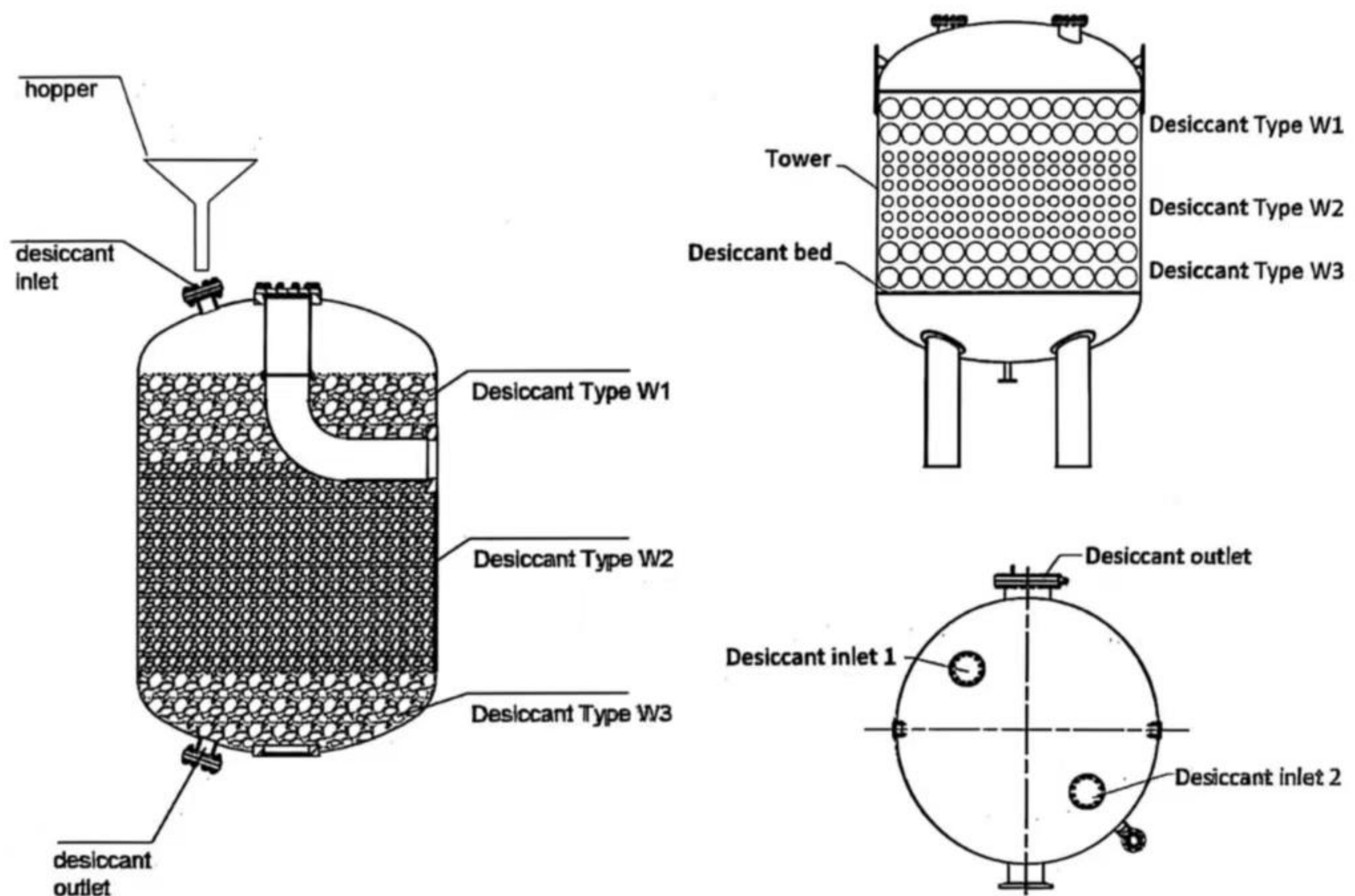
Detail: product that needs filling desiccant on site for each dryer vessel, as below:

Product model Standard	tower top W1 1624602851, KG	tower middle W2 1624602849, KG	tower bottom W3 1624602851, KG
XD2750 ZP G	250	2000	250
XD3600 ZP G&S	300	2300	300
XD4150 ZP G&S	350	2700	350
XD5000 ZP G&S	425	3250	425
X4150 P G&S	325	2550	325
X5000 P G&S	400	3050	400

Product model HD option	tower top W1 1639614016, KG	tower middle W2 1639614015, KG	tower bottom W3 1624602849, KG
XD2750 ZP G	500	1250	750
XD3600 ZP G&S	575	1450	875
XD4150 ZP G&S	675	1700	1025
XD5000 ZP G&S	825	2050	1225
X4150 P G&S	-	950	2250
X5000 P G&S	-	1150	2700

Product model	tower top W1 1624602851, KG	tower middle W2 1624602849, KG	tower bottom W3 1624602851, KG
XD6000 ZP G&S	520	4160	520
XD7500 ZP G&S	650	5200	650
XD9000 ZP G&S	780	6240	780
XD10000 ZP G&S	870	6960	870

5.2 filling sequence:



Filling sequence instruction: XD2750-5000&X4150-5000 (left) and XD6000-10000 (right)

Attention:

According to the weight of the desiccant and the height of the dryer and compressor room, it's important to select the suitable crane or tooling to fill the desiccant.

6 Filling Instruction

- (1) Filling the desiccant into the vessel through the desiccant inlet port on the top of the vessel. It is recommended to use a funnel.



Warning: It is NOT allowed to fill the desiccant under pressure by using a blowing filling system or any other forced filling system as this can shatter and damage the desiccant, Filling only by the desiccant itself gravity.

Do not overfill:

- (1) Level of desiccant should not be higher than top strainer lower support. See drawing above. Make sure strainer remains free of desiccant.
- (2) Isolate PDP sensor if installed
- (3) The amount of dust generated after filling can clog the prefilter (newer models) or PDP sensor element very quickly.
- (4) Open PDP shut-off valve after minimum one full cycle (two towers must have switched);
- (5) Verify optional pipe filters, if installed. It may be necessary to replace the dust filter cartridge(s) after 100-200 hours because of dust from filling. If no filters are installed, it is highly recommended to blow out the desiccant bed for a few minutes to remove dust. This must be done by blowing to atmosphere (use installed blower) to prevent that dust enters the compressed air piping and the customer's installation.
- (6) When heavy duty, we need to fill in with some silica gel. When we fill silica gel pay attention with water, keep silica gel away from water. If silica gel contacts with free water, then after a while the water starts boiling, so not all desiccants have this.

Treatment and disposal of waste desiccant:

Open the outlet port of the vessel to let the desiccant flow out. Once the desiccant is no longer flowing automatically, use a rake, hoe, shovel, or vacuum tank car to remove the rest of the desiccant. Never enter the vessel to remove any remaining desiccant. When disposing of used desiccant, be sure to discard it in such an

environmentally acceptable manner. If any problem involves of the desiccant disposal, please follow the local solid waste treatment and disposal law or regulation.

Since the desiccant outlet port is slightly higher than the welded support frame inside the vessel, it's important for users to empty the whole vessel.

The empty procedure should be:

- (1) Let the dryer program run to the pressure equalization step.
- (2) Stop the dryer when the pressure is equalized in both towers.
- (3) Shut off the power supply.
- (4) Close the pilot air valve. Isolate the dryer from the air net by closing the air inlet and outlet valves. Depressurize the towers.
- (5) Install a suitable collector underneath the plug for emptying; note that the desiccant may be very hot
- (6) Remove the plugs or flanges and let the desiccant pour out into the collector. A vacuum cleaner may be used to speed up the process and control dust.

At any time during the dumping process, if the desiccant start to burning or reacting (releasing a lot of heat), please stop dumping process. Evacuate all people from filling area to safety area. Well trained staff must be equipped with self-contained oxygen breathing equipment to return to the filling area to prevent desiccant reacting and burning. Use the extinguishing tap to submerge the desiccant outside the vessel to prevent reaction and fire. Contact desiccant supplier or Atlas Copco for assistance.

When dealing with fresh desiccant and used desiccant, never put it into the mouth, and contact with eyes and skin. When dumping desiccant, operators must wear gloves, eye shields, dust masks, safety shoes and protective clothing. Usually there is a desiccant bed welded within the bottom of the vessel. The welded bed is used to place the desiccant.

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7 Desiccant support frame inspection

Follow desiccant discharge procedure, once all the desiccant is removed, make sure the air quality inside the vessel is safe to enter.

The following precautions must be taken to prevent suffocation before entering the vessel:

1. A protection plate shall be installed in all inlet and outlet piping to prevent accidental discharge of dangerous fluids into vessel, in case desiccant flows out hurt people. Before operator enters, an inside air quality analysis must be done to ensure operator work inside the vessel can breathe safely.
2. Keep air well ventilated inside the vessel. The compressed air cannot be used for ventilation, since compressed air will easily be contaminated or filled by other gases due to faulty operation.
3. Only well trained and PPE equipped operator can be entered the vessel. Make sure operator always wear seat belts and safety ropes. Another two well trained and PPE equipped operators must keep an eye on the operators whose work inside the vessel.

Instruction 指导

1 General information 常规信息

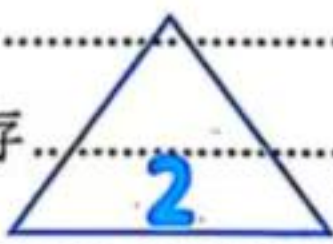
Desiccant filling on-site instruction 干燥剂填充现场指导

Applicable to 适用于:	Desiccant filling on-site	
Safety Instruction 安全指导:	General 常规	
Revision 版本:	Date 日期:	Modified from 改动:
1	07.03.2023	Heavy duty of 3600-4150 added Desiccant filling of 6000-10000 added
2	11.09.2023	Section "Package, Transport and Storage" added; Std & Heavy filling of XD2750 ZP G added; Desiccant ratio of 3600/4150/5000 modified;

2 Document Overview 文档概述

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3 Package, Transport and Storage 包装, 运输和储存

1. The desiccant shipping packages adopt double-layer packaging, the inner bag is a polyethylene film bag, and the outer bag is a plastic woven bag or a paper-plastic-composite packaging bag. The inner bag is tied or heat sealed, and the outer bag should be firmly sewn tightly.

If the user has special requirements, the supply and demand parties shall negotiate.

吸附剂装运包装采用双层包装, 内袋为聚乙烯薄膜袋, 外袋采用塑料编织袋或纸塑复合包装袋, 内袋扎口或热合封口, 外袋应牢固缝合。

用户如有特殊要求, 由供需双方协商。

2. During transportation and storage, the desiccant should be avoided from mixing with volatile substances, should be protected from rain and damage. The packaging should be handled with care to avoid contamination.

吸附剂在运输和储存过程中, 应避免与易挥发性物质混运, 防止雨淋, 不得受损, 包装应小心轻放, 避免污损。

3. The shipping packages with desiccant should be stored in a dry, enclosed warehouse. These packages must be protected from harm from nature and surroundings.

吸附剂装运包装应当储存在干燥、封闭的仓库里。必须保护吸附剂装运包装免受来自大自然和周围环境的侵害。

4. The shipping packages with desiccant must be stored in an area that is off the ground and well drained to prevent contact with accumulated water or mud.

吸附剂装运包装必须储存在离开地面且排水良好的区域, 防止接触到积水或泥巴。

5. The shipping packages should be covered with waterproof or oilcloth to prevent water accumulation or infiltration at the top.

吸附剂装运包装应该用防水布或油布盖住, 防止顶部积水或水分渗入。

6. The shipping packages with desiccant are not recommended for long-term storage outdoors. The impacts of outdoor storage on the desiccant are unknown.

吸附剂装运包装不推荐长期存放在室外。室外储存给吸附剂的影响未知。

7. If shipping packages with desiccant do need to be stored outdoors for a short period of time, measures should be taken to prevent rain, water, and moisture, as well as UV (ultraviolet) degradation.

如果吸附剂确实需要在室外短时间储存, 需采取措施防雨, 防水和防潮, 此外, 还要防止 UV (紫外线) 降解。

8. The shelf life of desiccant supplied through large bag packaging varies depending on the type of product, so please consult your local representative for the storage time. It is advised that desiccant shall be filled into the specific pressure vessel as soon as possible to protect desiccant and ensure its performance.

通过大袋包装供应的干燥剂的保质期因产品类型而异, 因此请咨询当地代表了解储存时间。建议尽快将干燥剂填充到特定的压力容器中, 以保护干燥剂并确保其性能。

9. Due to the strong adsorption capacity of desiccant, if the storage time is too long, the desiccant should absorb moisture from the air, which will affect the adsorption performance. If the situation is serious, the desiccant needs to be regenerated for several cycles before it can enter a good working state.

由于吸附剂吸附能力比较强，如储存时间过长，吸附剂吸附了空气中的水分，进而影响吸附效果，如情况严重，吸附剂需再生几个周期，方能进入良好工作状态。

10. Any damage or malfunction caused by not following the above instructions is not covered by Warranty or Product Liability.

因未遵守上述说明，而造成的吸附剂任何损坏或故障不在保修或产品责任范围内。

11. Package, transport and storage of desiccant should comply with relevant laws and local regulations. Activated alumina refers to HG/T 3927-2020. Silica gel refers to HG/T2765-2005. Molecular sieve refers to HG/T2524-2010. For more details, contact your local representative.

吸附剂的包装、运输和储存应符合相关法律和当地法规的规定。活性氧化铝参照 HG / T 3927-2020。硅胶参照 HG/T2765-2005。分子筛参照 HG/T2254-2010。有关更多详细信息，请联系您当地的代表。

4 Safety

4.1 Safety first

SAFETY = THINK BEFORE YOU ACT
安全=在行动之前思考



6. STOP 停止
EVALUATE RISKS 评估
风险



5. THINK 思考
SEEK PROPER SOLUTION
寻求适当解决方案



4. ACT 行动
CARRY OUT SOLUTION
实施解决方案

了解你客户的安全规则



KNOW THE SAFETY RULES OF YOUR CUSTOMER

了解你机器的安全指导说明书



KNOW THE SAFETY INSTRUCTIONS OF YOUR MACHINE



PERSONAL PROTECTION KEEPS YOU SAFE
个人保护措施保障你安全



LOCK OUT / TAG OUT 安全锁/安全标签
TAKE AWAY THE RISKS 消除风险

4.2 Prepare work 准备工作

Attention 注意: Protect yourself, protect others. Reading and understanding this section completely before filling the desiccant and pay attention to all warning labels for shipping packages.

保护自己，保护他人。在填充干燥剂之前阅读和完全理解本节并注意所有的装运包装警告标签。

When the desiccant is fresh and unused, it's property is not flammable. However, when it's exposed to water, it's temperature could be raised quite high. When the desiccant first time become wet, the temperature can reach the boiling point of water. Therefore, the desiccant should not be put in the mouth, or contact with the eyes.

当干燥剂是新的和未使用的,它的属性是不易燃的。然而,当接触到水,它的温度将会升得相当高。当干燥剂第一次变得潮湿,温度可以达到水的沸点。因此,干燥剂不应放在嘴里,或接触眼睛。

Desiccant dust may irritate your nose, throat, eyes, lungs and skin. As some desiccant dust will be produced when filling with fresh desiccant, it is necessary to wear protective glasses, dust mask, gloves and protective clothing.

干燥剂粉尘会刺激你的鼻子、喉咙、眼睛、肺部和皮肤。因为在灌装新鲜的干燥剂时将产生一些干燥剂粉尘,所以必须戴防护眼镜、防尘口罩、手套和防护服。

Desiccant is usually stored within a bag or barrel package with a specified weight. Pay attention to safety when opening the cover of the bag package, you must wear protective glasses, dust mask, gloves and protective clothing for protection.

干燥剂通常是存储在一个指定的重量的包或桶里。在打开包的盖子时要注意安全,你必须戴防护眼镜、防尘口罩、防护手套和防护服。

Since desiccant filling port is at the top of the vessel, working at high altitude above ground is needed, therefore local climbing operation safety regulations must be observed.

因为干燥剂灌装口在容器顶部,所以需要在高空地面作业,因此必须遵守当地攀爬操作安全规定。

4.2.1 Evacuation of desiccant filling area 撤空干燥剂填充区域

Due to all dryers desiccant filling port are at the top of the vessel, the desiccant must be lifted to the top of the vessel by the crane, therefore operating area of the crane must be identified, all personnel must be kept away from the crane operation area and stand within the safety area. Sometimes a pneumatic filling system is used to blow the desiccant into the vessel through the pipe connection. The inlet and outlet of the pipe should be kept away from people because of the air pressure in the storage tank. As the desiccant passes through the system, the transportation and receiving system must be well grounded to eliminate the static charge. If flammable vapors, such as hydrocarbons, are present in the filling area, well grounding is extremely important to avoid electrostatic explosions and fire.

由于所有干燥剂加注口都在容器顶部,干燥剂必须由起重机提升到容器顶部,因此必须确定起重机的操作区域,所有人员必须远离起重机操作区域并站在安全区域内。有时使用气动加注系统通过管道连接将干燥剂吹入容器。由于储罐内的空气压力,管道进出口应远离人员。干燥剂通过系统时,运输和接收系统必须接地良好,以消除静电。如果填充区域存在易燃蒸汽,如碳氢化合物,则良好的接地对于避免静电爆炸和火灾极其重要。

4.2.2 Personnel protection equipment (PPE) 人员保护设备

Operator must wear safety belts and ropes when entering the vessel.

进入容器时操作员必须系好安全带和绳索。

If operators must enter the vessel, the following precautions must be taken to prevent suffocation:

如果操作人员必须进入容器，必须采取以下预防措施防止窒息：

1. A protection plate shall be installed in all inlet and outlet piping to prevent accidental discharge of dangerous fluids into vessel. Before operator enters, an inside air quality analysis must be done to ensure operator work inside the vessel can breathe safely.
应在所有入口和出口管道上安装保护板，以防止危险液体意外排放到容器中。操作人员进入前，必须进行内部空气质量分析，以确保操作人员在容器内工作时能够安全呼吸；
2. Keep air well ventilated inside the vessel. The compressed air cannot be used for ventilation, since compressed air will easily be contaminated or filled by other gases due to faulty operation.
保持容器内通风良好。压缩空气不能用于通风，因为压缩空气很容易受到污染或因操作不当而被其他气体填充；
3. Operator who works at high altitude must wear suitable PPE and observe local climbing operation safety regulation.
高空作业人员必须穿戴合适的劳动防护用品，遵守当地登高作业安全规定；

Only well trained and PPE equipped operator can be entered the vessel. Make sure operator always wear seat belts and safety ropes. Another two well trained and PPE equipped operators must keep an eye on the operators whom work inside the vessel.

只有训练有素并配备劳动防护用品的操作人员才能进入容器。确保操作员始终系好安全带和安全绳。另外两名训练有素且配备 PPE 的操作人员必须密切关注在容器内工作的操作人员。

4.2.3 Safety requirement 安全要求

(1) Wear PPE 佩戴 PPE

Gloves, Protective glasses, Dust Mask, Safety shoes, Safety vest.

手套、防护眼镜、防尘口罩、安全鞋、安全背心。

Warning 警告: Operators must wear safety ropes and seat belts when working high above the ground. 高空作业时，操作人员必须系好安全绳和安全带。

- (2) Avoid the dusting of desiccant particles. 避免干燥剂颗粒扬尘；
 - (a) Slowly open the desiccant transportation package. 缓慢打开干燥剂运输包装；
 - (b) When the desiccant is filled by a pneumatic filling system, the outlet shall not toward to the operator.
当干燥剂由气动加注系统加注时，出口不得朝向操作员。
- (3) Eliminate the potential risk of the desiccant bag fallen off – nobody can walk into the crane lifting area.
消除干燥剂袋脱落的潜在风险-任何人不得进入起重机起重区域。
Before working on lift the desiccant bag by crane, operating area of the crane must be identified, all personnel must be kept away from the crane operation area and stand within the safety area.
用起重机吊起干燥剂袋前，必须确定起重机的操作区域，所有人员必须远离起重机操作区域，站在安全区域内；
- (4) If the desiccant is filling by a pneumatic filling system, avoid flammable steam burning, pneumatic filling system must be well grounded to avoid static electricity and cause fire. A static spark can ignite any flammable vapor at the site.
如果干燥剂是通过气动加注系统加注的，应避免可燃蒸汽燃烧，气动加注系统必须接地良好，以防静电和引起火灾。静电火花可点燃现场的任何易燃蒸汽；
- (5) Avoid the possibility of suffocation and falling before entering the vessel or working at the top of the vessel.
在进入容器或在容器顶部工作前，避免窒息和坠落的可能性

- (a) Install baffles at the inlet and outlet of the vessel. 在容器进出口安装挡板
- (b) Measure the air quality inside the vessel to ensure safety access.
测量容器内的空气质量，确保安全进入。
- (c) Operator must be well-trained and PPE equipped, also safety rope is needed.
操作人员必须经过良好培训，并配备个人防护用品，还需要安全绳。
- (d) Another two well trained and fully PPE equipped operator must be cooperate with the one whom will work inside the vessel or work at the top of the vessel.
另外两名训练有素且配备齐全的 PPE 操作员必须与将在容器内工作或在容器顶部工作的操作员合作。

5 Information of filling on-site dryer 填装现场干燥机信息

5.1: Machine type above XD3600 or total weight (with or without desiccant) over 15000kg, it's required to ship out the machine without desiccant, due to safety reason during the transportation. Filling desiccant can only start once the machine has been located at the correct position at customer site.

XD3600 以上机型或总重量（含或不含干燥剂）15000kg 以上，运输过程中由于安全原因，要求不含干燥剂装运。只有当机器位于客户现场的正确位置时，才能开始加注干燥剂。

Detail: product that needs filling desiccant on site for each dryer vessel, as below:

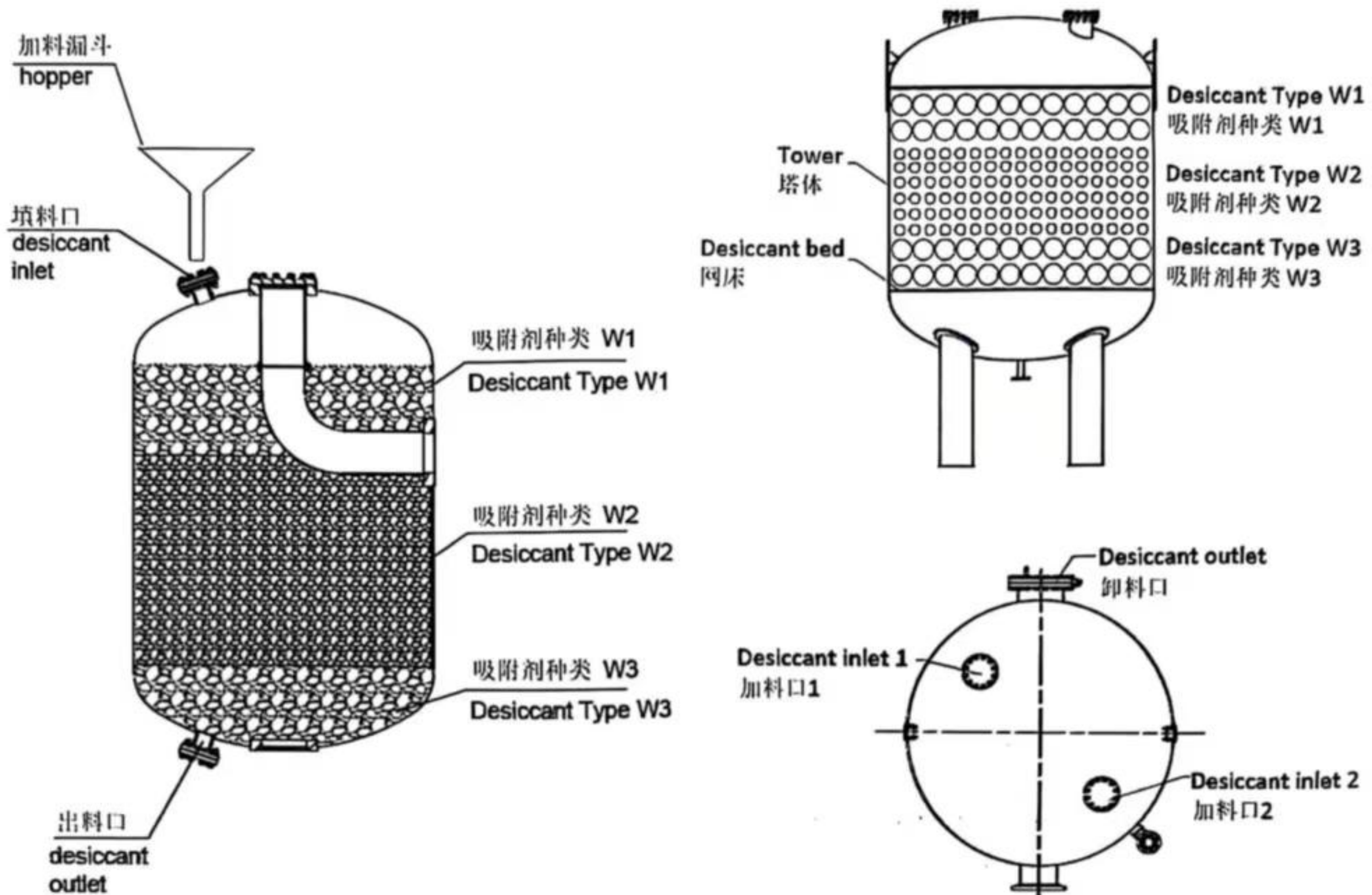
现场每个干燥塔需要填充的干燥剂产品，详情如下：

Product model Standard	tower top W1 1624602851, KG	tower middle W2 1624602849, KG	tower bottom W3 1624602851, KG
XD2750 ZP G	250	2000	250
XD3600 ZP G&S	300	2300	300
XD4150 ZP G&S	350	2700	350
XD5000 ZP G&S	425	3250	425
X4150 P G&S	325	2550	325
X5000 P G&S	400	3050	400

Product model HD option	tower top W1 1639614016, KG	tower middle W2 1639614015, KG	tower bottom W3 1624602849, KG
XD2750 ZP G	500	1250	750
XD3600 ZP G&S	575	1450	875
XD4150 ZP G&S	675	1700	1025
XD5000 ZP G&S	825	2050	1225
X4150 P G&S	-	950	2250
X5000 P G&S	-	1150	2700

Product model	tower top W1 1624602851, KG	tower middle W2 1624602849, KG	tower bottom W3 1624602851, KG
XD6000 ZP G&S	520	4160	520
XD7500 ZP G&S	650	5200	650
XD9000 ZP G&S	780	6240	780
XD10000 ZP G&S	870	6960	870

5.2 filling sequence: 填充顺序



Filling sequence instruction: XD2750-5000&X4150-5000 (left) and XD6000-10000 (right)

加料顺序说明: XD2750-5000&X4150-5000 (左) 和 XD6000-10000 (右)

Attention 注意:

According to the weight of the desiccant and the height of the dryer and compressor room, it's important to select the suitable crane or tooling to fill the desiccant. 根据干燥剂的重量以及干燥剂和压缩机房的高度，选择合适的起重机或工具来填充干燥剂非常重要。

6 Filling Instruction 灌装指导

- (1) Filling the desiccant into the vessel through the desiccant inlet port on the top of the vessel. It is recommended to use a funnel. 通过容器顶部的干燥剂入口将干燥剂注入容器，建议使用漏斗。



Warning: It is NOT allowed to fill the desiccant under pressure by using a blowing filling system or any other forced filling system as this can shatter and damage the desiccant, Filling only by the desiccant itself gravity.

不允许使用吹气填充系统或任何其他强制填充系统在压力下填充干燥剂，因为这可能会粉碎和损坏干燥剂，仅通过干燥剂本身的重力填充。

Do not overfill: 不要过量加注：

- (1) Level of desiccant should not be higher than top strainer lower support. See drawing above. Make sure strainer remains free of desiccant. 干燥剂的高度不应高于顶部过滤器下部支架。见上图。确保滤网没有干燥剂。
- (2) Isolate PDP sensor if installed 隔离 PDP 传感器（如已安装）
- (3) The amount of dust generated after filling can clog the prefilter (newer models) or PDP sensor element very quickly. 填充后产生的灰尘量会很快堵塞前置过滤器（新型号）或 PDP 传感器元件。
- (4) Open PDP shut-off valve after minimum one full cycle (two towers must have switched); 至少一个完整循环后打开 PDP 切断阀（两个塔必须已切换）；
- (5) Verify optional pipe filters, if installed. It may be necessary to replace the dust filter cartridge(s) after 100-200 hours because of dust from filling. If no filters are installed, it is highly recommended to blow out the desiccant bed for a few minutes to remove dust. This must be done by blowing to atmosphere (use installed blower) to prevent that dust enters the compressed air piping and the customer's installation.
验证可选管道过滤器（如已安装）。可能需要在 100-200 小时后更换滤尘器筒，因为填充时会产生灰尘。如果没有安装过滤器，强烈建议将干燥剂层吹几分钟以除去灰尘。这必须通过向大气吹气（使用已安装的鼓风机）来完成，以防止灰尘进入压缩空气管道和客户的装置。
- (6) When heavy duty, we need to fill in with some silica gel. When we fill silica gel pay attention with water, keep silica gel away from water. If silica gel contacts with free water, then after a while the water starts boiling, so not all desiccants have this.

当重负载时，我们需要用一些硅胶填充。当我们填充硅胶时，注意用水，保持硅胶远离水。如果硅胶与自由水接触，那么过一会儿水就开始沸腾，所以不是所有的干燥剂都有这种情况。

Treatment and disposal of waste desiccant 废干燥剂的处理与处置

Open the outlet port of the vessel to let the desiccant flow out. Once the desiccant is no longer flowing automatically, use a rake, hoe, shovel, or vacuum tank car to remove the rest of the desiccant. Never enter the vessel to remove any remaining desiccant. When disposing of used desiccant, be sure to discard it in such an environmentally acceptable manner. If any problem involves of the desiccant disposal, please follow the local solid waste treatment and disposal law or regulation

打开容器出口，让干燥剂流出。一旦干燥剂不再自动流动，使用耙子、锄头、铲子或真空罐车清除干燥剂的其余部分。切勿进入容器清除任何残留干燥剂。处理用过的干燥剂时，务必以环境可接受的方式丢弃。如果干燥剂处理有任何问题，请遵守当地固体废物处理和处置法律或法规。

Since the desiccant outlet port is slightly higher than the welded support frame inside the vessel, it's important for users to empty the whole vessel. 由于干燥剂出口略高于容器内焊接支撑架，因此用户清空整个容器非常重要。

The empty procedure should be: 清空过程应如下：

- (1) Let the dryer program run to the pressure equalization step. 让干燥机程序运行到压力均衡步骤。
- (2) Stop the dryer when the pressure is equalized in both towers. 两塔均压后，停止干燥机。
- (3) Shut off the power supply. 切断电源。
- (4) Close the pilot air valve. Isolate the dryer from the air net by closing the air inlet and outlet valves. Depressurize the towers. 关闭导向气阀门。关闭进气阀和排气阀，将干燥器与空气网隔离。降低塔的压力。
- (5) Install a suitable collector underneath the plug for emptying; note that the desiccant may be very hot in the plug. 在堵头下面安装一个合适的收集器，以便排空；注意干燥剂可能非常热。
- (6) Remove the plugs or flanges and let the desiccant pour out into the collector. A vacuum cleaner may be used to speed up the process and control dust. 拆下堵头或法兰，让干燥剂倒入收集器中。真空吸尘器可用于加速处理和控制灰尘。

At any time during the dumping process, if the desiccant start to burning or reacting (releasing a lot of heat), please stop dumping process. Evacuate all people from filling area to safety area. Well trained staff must be equipped with self-contained oxygen breathing equipment to return to the filling area to prevent desiccant reacting and burning. Use the extinguishing tap to submerge the desiccant outside the vessel to prevent reaction and fire. Contact desiccant supplier or Atlas Copco for assistance.

在倾倒过程中的任何时候，如果干燥剂开始燃烧或反应（释放大量热量），请停止倾倒过程。将所有人员从灌装区疏散到安全区。训练有素的员工必须配备自给式氧气呼吸设备，以返回灌装区，防止干燥剂反应和燃烧。用灭火水龙头将干燥剂浸入容器外，防止反应和火灾。联系干燥剂供应商或 Atlas Copco 寻求帮助。

When dealing with fresh desiccant and used desiccant, never put it into the mouth, and contact with eyes and skin. When dumping desiccant, operators must wear gloves, eye shields, dust masks, safety shoes and protective clothing. Usually there is a desiccant bed welded within the bottom of the vessel. The welded bed is used to place the desiccant.

处理新鲜干燥剂和用过的干燥剂时，切勿将其放入口中，并与眼睛和皮肤接触。倾倒干燥剂时，操作人员必须戴手套、护眼板、防尘口罩、安全鞋和防护服。通常在容器底部焊接有干燥剂网床。焊接网床适用于放置干燥剂。

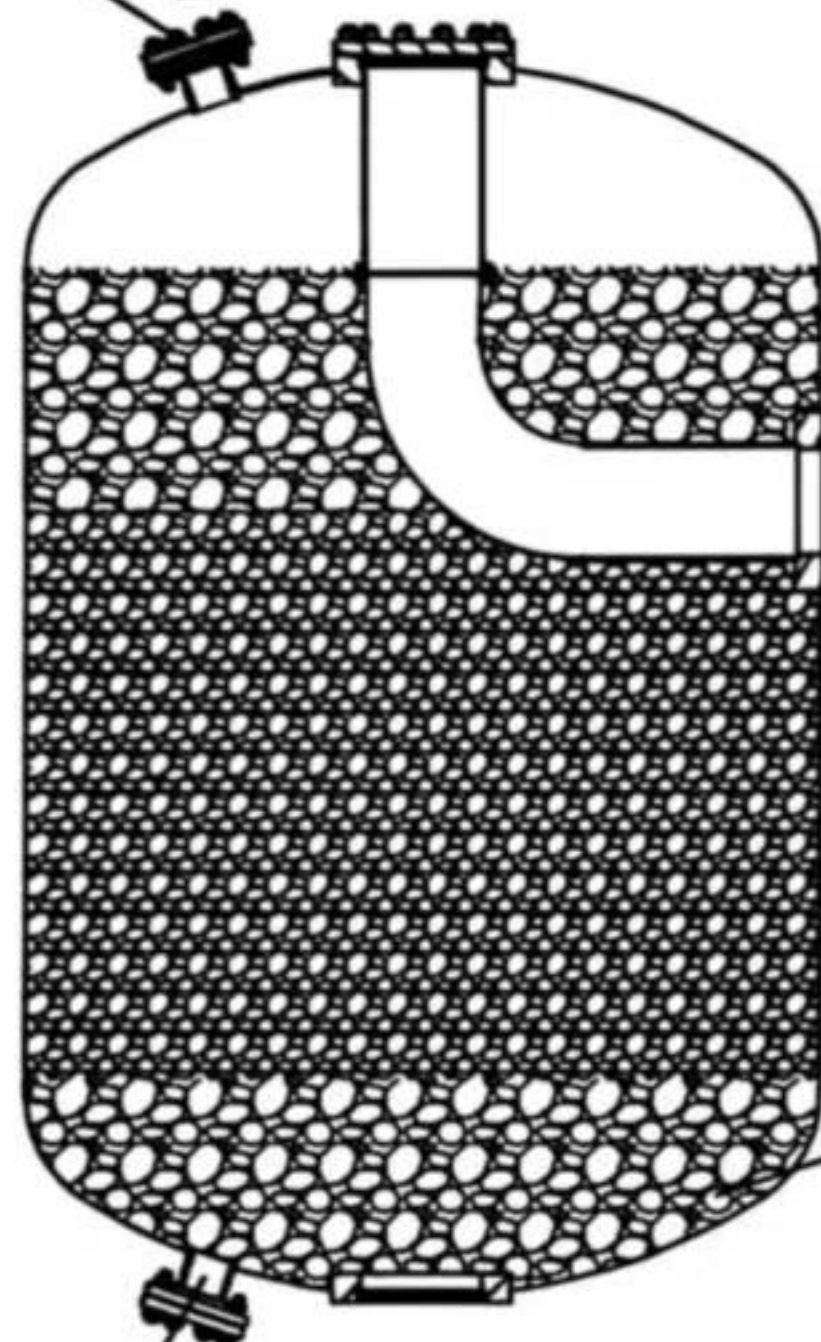
7 Desiccant support frame inspection 干燥剂支撑架检查

Follow desiccant discharge procedure, once all the desiccant is removed, make sure the air quality inside the vessel is safe to enter. 遵循干燥剂排放步骤，一旦去除所有干燥剂，确保容器内的空气质量是可以安全进入的。

The following precautions must be taken to prevent suffocation before entering the vessel:

在进入容器之前，必须采取以下预防措施防止窒息：

1. A protection plate shall be installed in all inlet and outlet piping to prevent accidental discharge of dangerous fluids into vessel, in case desiccant flows out hurt people. Before operator enters, an inside air quality analysis must be done to ensure operator work inside the vessel can breathe safely.
应在所有入口和出口管道上安装保护板，以防止危险液体意外排放到容器中，以防干燥剂流出伤人。操作人员进入前，必须进行内部空气质量分析，确保操作人员在容器内工作时能够安全呼吸；
2. Keep air well ventilated inside the vessel. The compressed air cannot be used for ventilation, since compressed air will easily be contaminated or filled by other gases due to faulty operation.
保持容器内通风良好。压缩空气不能用于通风，因为压缩空气很容易受到污染或因操作不当而被其他气体填充；
3. Only well trained and PPE equipped operator can be entered the vessel. Make sure operator always wear seat belts and safety ropes. Another two well trained and PPE equipped operators must keep an eye on the operators who work inside the vessel.
只有训练有素、配备 PPE 的操作人员才能进入容器。确保操作员始终系好安全带和安全绳。另外两名训练有素且配备 PPE 的操作人员必须密切关注在容器内工作的操作人员。

加料漏斗
hopper填料口
desiccant
inlet

吸附剂种类 W1

Desiccant Type W1

吸附剂种类 W2

Desiccant Type W2

吸附剂种类 W3

Desiccant Type W3

出料口
desiccant
outlet

注：此示意图是为了标识装填干燥剂的顺序及用量（每个干燥塔）
详细的干燥剂装填注意事项及方法详见：
干燥剂现场装填操作说明

Note: this drawing is to indicate the order sequence
and amount of desiccant filling (for each dryer tower)
Operate detailed in:
Desiccant filling on-site instruction

Product model	tower top W1	tower middle W2	tower bottom W3
Standard	1624602851, KG	1624602849, KG	1624602851, KG
XD2750 ZP G	250	2000	250
XD3600 ZP G&S	300	2300	300
XD4150 ZP G&S	350	2700	350
XD5000 ZP G&S	425	3250	425
X4150 P G&S	325	2550	325
X5000 P G&S	400	3050	400

Product model	tower top W1	tower middle W2	tower bottom W3
HD option	1639614016, KG	1639614015, KG	1624602849, KG
XD2750 ZP G	500	1250	750
XD3600 ZP G&S	575	1450	875
XD4150 ZP G&S	675	1700	1025
XD5000 ZP G&S	825	2050	1225
X4150 P G&S	-	950	2250
X5000 P G&S	-	1150	2700

Product model	tower top W1	tower middle W2	tower bottom W3
	1624602851, KG	1624602849, KG	1624602851, KG
XD6000 ZP G&S	520	4160	520
XD7500 ZP G&S	650	5200	650
XD9000 ZP G&S	780	6240	780
XD10000 ZP G&S	870	6960	870

All materials supplied are in compliance with the
requirements of the List of Prohibited Substances

Tolerances, if not indicated, according to					
ATLAS COPCO STANDARD/Klasse/Class					
Name 干燥剂现场装填示意图 (AC)		desiccant filling on-site drawing		Security Class	
Material				1102 K/ 2	
Treatment				ACD	
Scale		Family		A3	
Drawn by		Blank nr.		Replaces	
Version Dwg		Blank wt.		A/g gov.	
Des checked		Prod checked		Approved	
New				Date	
				Designation	
				Sheet () 01 01	
				9829226071	

02.00		Update desiccant ratios of 3600-6000 and add desiccant ratios of 2750	Mar/07/2023						
Note	Posi- tion	Modified from	Date	Int./Appd.	Parent 3D Model	Ed./Version 3D			